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# Bihar Engineering University, Patna

## B.Tech 5th Semester Exam., 2020

*Time: 3 hours*

*Code: 101503* *Subject: Geotechnical Engineering - I* *Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are NINE questions in this paper.*
*(iii) Attempt FIVE questions in all.*
*(iv) Question No. 1 is compulsory.*

***Q.1 Choose the correct answer of the following (any seven) :*** **[14]**

(a) Water transported soils are termed as
(i) alluvial (iii) talus
(ii) aeoline (iv) loess

(b) The three-phase soil system consists of
(i) soil, water and air (iii) solids, water and air
(ii) soil, semi-solid and air (iv) solids, semi-solids and water

(c) The standard and most accurate method for determination of water content is
(i) calcium carbide method (iii) oven drying method
(ii) torsion balance method (iv) alcohol method

(d) The property of fine grained soils to change from liquid state to plastic state with decrease of water content is known as
(i) liquidity (iii) shrinkage
(ii) plasticity (iv) consistency

(e) The meniscus and dispersing agent corrections to hydrometer reading are respectively
(i) positive and positive (iii) negative and negative
(ii) positive and negative (iv) negative and positive

(f) The maximum particle size for which Darcy's law is applicable is
(i) 75 μm (iii) 0.5 mm
(ii) 0.2 mm (iv) 0.1 mm

(g) A soil has a discharge velocity of $6\times10^{-6}$m/s and the porosity of 0.40. Its seepage velocity is
(i) $15\times10^{-6}$m/s (iii) $2.4\times10^{-6}$m/s
(ii) $1.5\times10^{-6}$m/s (iv) $6\times10^{-6}$m/s

(h) If water rises (unit weight of water = $\gamma_w$) by a height h above the groundwater table by capillary action, the pore water pressure the height at h above groundwater table is
(i) zero (iii) $-\gamma_w h$
(ii) $\gamma_w h$ (iv) $\gamma_w h^2$

(i) The number of blows used for compacting each layer in modified proctor test is
(i) 25 blows (iii) 55 blows
(ii) 56 blows (iv) 76 blows

(j) In which of the following footings of equal width, the vertical stress extends to larger depth below its base?
(i) Square footing (iii) Circular footing
(ii) Strip footing (iv) Rectangular footing

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***Q.2 Solve both questions :*** **[14]**

(a) Distinguish clearly between physical and chemical weathering.

(b) A saturated clay has a water content of 40% and a bulk specific gravity of 1.75. Determine the void ratio and specific gravity of particles. [Take unit weight of water = 10 kN/m³]

***Q.3 Solve both questions :*** **[14]**

(a) Write a short note on the 'method of determination of Atterberg limits'.

(b) A fully saturated clay has a water content 40% and a mass specific gravity of 1.92. After oven drying, the mass specific gravity reduces to 1.81. Determine the specific gravity of soil solids and shrinkage limit.

***Q.4 Solve both questions :*** **[14]**

(a) Sketch the grain size distribution curve for uniformly graded, well-graded and gap-graded soils and discuss their Characteristics.

(b) Two soils $S_1$ and $S_2$ are tested in the laboratory for the consistency limits. The data available is as follows :

| | **Soil S1** | **Soil S2** |
|---|---|---|
| Plastic limit | 21% | 31% |
| Liquid limit | 61% | 65% |
| Flow index | 10 | 5 |
| Natural moisture content | 40% | 50% |

(i) Which one is more plastic? (ii) Which one is better foundation material when remoulded? (iii) Which one has better strength as a function of water content? (iv) Which one has better strength at the plastic limit?

***Q.5 Solve both questions :*** **[14]**

(a) How do you determine the permeability of cohesive soils in the laboratory? Derive the formula you used.

(b) A field pumping out test gave the following data : Diameter of well = 30 cm; Thickness of confined aquifer = 35 m; Radius of circle of influence = 300 m; Drawdown during the test = 5.5 m; Pump discharge = 0.9 cumecs. Compute the permeability of the aquifer.

***Q.6 Solve both questions :*** **[14]**

(a) What is quick sand? How would you calculate the hydraulic gradient required to create quick sand conditions in sand?

(b) A sand deposit consists of two layers. The top layer is 3.0 m thick (unit weight = 17 kN/m³) and the bottom layer is 4.0 m thick (unit weight = 18 kN/m³). The water table is at a depth of 4.0 m from the top. Draw the diagram and show the variation of total stress, neutral stress and effective stress.

***Q.7 Solve both questions :*** **[14]**

(a) What are the factors that affect compaction? Discuss in detail.

(b) A laboratory compaction test on the soil having specific gravity 2.70 gave an MDD of 1.85 gm/cc and a water content of 19%. Determine the degree of saturation, air content and percentage air voids at the MDD. What would be theoretical MDD corresponding to zero air voids at the OMC?

***Q.8 Solve both questions :*** **[14]**

(a) Derive the equation for vertical stress due to line load of intensity q.

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(b) A point load of 250 kN acts on the ground surface. Using the Boussinesq analysis, find the maximum vertical pressure on the vertical plane at a distance of 3 m from the loading.

***Q.9 Write short notes on any four of the following:*** **[14]**

(a) Pheratic line
(b) Stress isobar
(c) Boiling sand condition
(d) Grain size analysis
(e) Sand replacement method
(f) Flow and toughness indices

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